# Task-Relative Descriptive Privilege in Linear Gaussian Dynamical Systems

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March 2026 *Preprint — not peer reviewed*

**Abstract**

We ask whether a single description can be simultaneously optimal for all prediction tasks in a linear dynamical system. In a finite-dimensional linear Gaussian dynamical system, the optimal rank-$r$ observer for a squared-error prediction target is uniquely determined by the top-$r$ eigenspace of a target information matrix derived from the stationary covariance and the dynamics. Two targets whose information matrices have distinct top-$r$ eigenspaces admit no jointly optimal observer. Global privilege holds if and only if the target family is *$r$-coherent*: all information matrices share a common top-$r$ eigenspace. $r$-coherence requires strong structural alignment between the dynamics and the task family. Outside it, privilege is task-relative within this class. We illustrate the theorem numerically: a concrete no-privilege instance yields a global-privilege gap of $\Gamma = 0.026 > 0$, and structural coherence is confirmed to be perturbation-robust.

**Keywords.** Linear Gaussian dynamical system; Ky Fan maximum principle; Grassmannian; eigenspace optimality; $r$-coherence; prediction; observer.

**MSC 2020.** 15A42 (primary); 93E11, 60G35 (secondary).

## 1 Introduction

A basic question in the study of dynamical systems is whether there is a single low-dimensional description that is optimal for every prediction task, or whether the optimal description depends on the target one seeks to predict. This paper answers that question exactly for finite-dimensional linear Gaussian dynamical systems (LGDS). The optimal rank-$r$ observer is the top-$r$ eigenspace of a target information matrix. Two targets whose information matrices have distinct dominant subspaces cannot be simultaneously optimised by any single observer. The exception has an exact algebraic characterisation: $r$-coherence of the target family.

The linear state-space framework is classical [6]; the relevant matrix optimisation on the Grassmannian [2] is governed by the Ky Fan maximum principle [3].

This yields three things useful for the present question: an exact risk formula with no approximation, a proof that task-relative privilege is the rule within this class, and an explicit algebraic criterion for when global privilege is warranted.

This result is intentionally narrow. It concerns only finite-dimensional linear Gaussian dynamical systems, fixed-rank linear observers, and squared-error prediction targets.

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## 2 System and Observer Family

**Dynamics.** Let

$$x_{t+1} = Ax_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \tag{1}$$

with $A$ stable (spectral radius $< 1$) and $Q \succ 0$. The unique stationary covariance $\Sigma$ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite [6].

**Observer family.** An observer of rank $r$ ($1 \le r < n$) is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r\times n}$, $UU^\top = I_r$:

$$\phi_U(x) = U\Sigma^{-1/2}x. \tag{2}$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing; in the static case ($A = I$, $\tau = 0$) this reduces to principal component analysis [5]. Two observers differing only by an orthogonal rotation of their row spaces are equivalent; the observer space is the Grassmannian $\mathrm{Gr}(r, n)$, a smooth compact Riemannian manifold of dimension $r(n - r)$, with metric and geodesic structure from $O(n)$ [2].

**Target family.** A prediction target is a pair $(C, \tau)$ with $C \in \mathbb{R}^{p\times n}$ and integer $\tau \ge 1$:

$$y_{C,\tau} = Cx_{t+\tau}. \tag{3}$$

Performance is total mean squared prediction error. A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is any finite collection.

## 3 Theorem

### 3.1 Exact Bayes risk formula

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. At stationarity, $\mathrm{Cov}(Cx_{t+\tau},\, z^\top) = CA^\tau\Sigma^{1/2}U^\top$ and $\mathrm{Cov}(z, z^\top) = I_r$. The Bayes risk is

$$\mathcal{R}(U;\, C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}\big(U\, M_{C,\tau}\, U^\top\big), \tag{4}$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top CA^\tau\Sigma^{1/2} \;\in\; \mathbb{R}^{n\times n}. \tag{5}$$

$M_{C,\tau}$ is symmetric positive semidefinite of rank at most $\min(p, n)$. The formula is exact.

### 3.2 Top-eigenspace optimality

**Proposition 1.** *For fixed rank $r$, the observer minimising $\mathcal{R}(U;\, C, \tau)$ over $\mathrm{Gr}(r, n)$ has row space equal to the top-$r$ eigenspace of $M_{C,\tau}$. The minimum risk is*

$$\mathcal{R}^* = \mathrm{tr}(C\Sigma C^\top) - \sum_{i=1}^{r} \lambda_i(M_{C,\tau}), \tag{6}$$

*where $\lambda_1 \ge \cdots \ge \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness up to rotation holds when $\lambda_r > \lambda_{r+1}$.*

*Proof.* Minimising $\mathcal{R}$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle [3, 4] the maximum equals $\sum_{i=1}^{r} \lambda_i(M_{C,\tau})$, achieved exactly when the row space of $U$ spans the corresponding eigenspace. □

### 3.3 No-global-privilege corollary

**Corollary 2.** *If targets $(C_1, \tau_1)$ and $(C_2, \tau_2)$ have information matrices $M_1$ and $M_2$ with distinct top-$r$ eigenspaces, no single rank-$r$ observer is simultaneously optimal for both.*

*Proof.* Optimality for target $k$ requires the row space of $U$ to equal the top-$r$ eigenspace of $M_k$. Distinct eigenspaces cannot be spanned by the same $U$. □

The corollary extends to any finite target family: if $\{M_k\}$ do not share a common top-$r$ eigenspace, no rank-$r$ observer is globally optimal.

### 3.4 $r$-coherence: the exact exception

**Definition 3.** *A target family $\mathcal{T}$ is* $r$-coherent *with respect to $(A, \Sigma)$ if all matrices $\{M_{C,\tau} : (C, \tau) \in \mathcal{T}\}$ share a common top-$r$ eigenspace $V^*$.*

**Corollary 4.** *Global privilege holds if and only if $\mathcal{T}$ is $r$-coherent.*

$r$-coherence requires all targets to be linear functionals of a common $r$-dimensional sufficient statistic across all horizons – a strong structural alignment between the dynamics and the task family. Outside $r$-coherence, the optimal observer generally depends on the target within the class studied here.

**Remark 5.** *$r$-coherence is an algebraic equality between eigenspaces, not an approximate or asymptotic condition. It is the exact characterisation of when global privilege is warranted.*

## 4 Numerical Illustrations

The theorem is analytically exact under its stated assumptions. We illustrate it numerically with two finite-dimensional instances: a no-global-privilege example and a structurally coherent example. All reported quantities are computed exactly from the analytic Bayes risk formula and numerical optimization on the Grassmannian.

**Setup.** State dimension $n = 8$, observer rank $r = 2$. The model is a finite-dimensional linear Gaussian dynamical system [7]. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$, random non-orthogonal $V$ (seed 100), $Q = 0.1\, I_8$. $\Sigma$ is computed from the discrete Lyapunov equation; $\Sigma \succ 0$ is confirmed to machine precision. All reported quantities are computed exactly from the analytic Bayes risk formula, eigenspace calculations, and repeated projected optimization over $\mathrm{Gr}(2, 8)$.

**Condition A: no-global-privilege instance.** Targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$.

With non-normal $A$, $A^1$ and $A^5$ rotate state-space directions differently; the principal angles [1] between the top-2 eigenspaces of $M_1$ and $M_2$ are $54.1°$ and $24.0°$, confirming that no single rank-2 observer is simultaneously optimal.

The global-privilege gap

$$\Gamma = \min_{U \in \mathrm{Gr}(r,n)} \max_k \left[\mathcal{R}(U; C_k, \tau_k) - \mathcal{R}_k^*\right] \tag{7}$$

Table 1: Regret matrix, Condition A ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

| Observer | Regret on $T_1$ | Regret on $T_2$ |
|---|---|---|
| $U_1^*$ (optimal for $T_1$) | 0.000 | 0.036 |
| $U_2^*$ (optimal for $T_2$) | 0.763 | 0.000 |
| $U_{\text{joint}}$ (compromise) | 0.026 | 0.026 |

measures the best achievable worst-case regret under task conflict. The minimax observer $U_{\text{joint}}$, found by repeated projected optimization over $\mathrm{Gr}(2, 8)$ from random initializations, achieves $\Gamma = 0.026 > 0$. No globally optimal observer exists for this target family. Table 1 gives the full regret structure. The asymmetry in the off-diagonal entries reflects the geometry of the instance: $U_2^*$ is optimised for directions nearly orthogonal to $T_1$, so its cross-regret is large; $U_1^*$ retains partial overlap with the long-horizon $T_2$ target, so its cross-regret is small. Both are computed from the same analytic formula.

**Condition B′: structural coherence and its robustness.** A target family is constructed so that all $M_k$ share the same top-2 eigenspace by design: normal $A_{\text{sym}} = VDV^\top$ with orthogonal $V$, and a common $C$ whose rows are the top-2 eigenvectors of $A$, varying only the prediction horizon $\tau \in \{1, 2, 3, 5\}$. Under this construction $M_{C,\tau}$ has the same dominant 2-dimensional eigenspace for all $\tau$, so the four optimal observers are identical by the theorem. The maximum pairwise principal angle across all four is 0.000° to machine precision.

Under $\varepsilon = 0.1$ perturbations to $C$ (20 random draws), the maximum principal angle across all draws and all pairs is 0.102 rad – well below the fragility threshold of 0.15 rad. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences of eigenspaces that break under small perturbations.

The numerical results are consistent with the theorem in all cases.

## 5 Scope

**What is earned.** In finite-dimensional linear Gaussian dynamical systems at stationarity, among rank-$r$ linear observers $\phi_U(x) = U\Sigma^{-1/2}x$ and squared-error prediction targets $y = Cx_{t+\tau}$, descriptive privilege is task-relative unless the target family is $r$-coherent. $r$-coherence requires strong structural alignment between dynamics and tasks. Outside it, no fixed-rank observer is globally optimal, and the cost of forcing any observer onto a non-native target is given exactly by the Bayes risk formula. The Grassmannian $\mathrm{Gr}(r, n)$ is the literal domain of this optimisation: a compact Riemannian manifold on which observer quality is a smooth function, with $O(n)$ acting by transformation.

**What is not earned.** This result does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or non-prediction tasks. The causal axis – whether the observer optimal for intervention differs from the observer optimal for prediction – is not addressed here.

## Acknowledgments

No external funding was received for this work. Code to reproduce all numerical illustrations is available at https://github.com/kunalb541/LGDS.

## References

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